Assignment 1: Store Sales Analysis & Prediction

1. Objective

The objective of this project was to analyse one year (366 days) of daily store sales data (January 1 – December 31, 2024), identify key revenue drivers, build a time-series forecasting model, and generate reliable short-term predictions. The focus was on combining technical modelling with actionable business insights.

2. Data Analysis & Key Findings

The dataset included three main variables: **Sales (target), Promotion (0/1), and DayOfWeek (1–7)**. The Date column was converted to datetime format and set as the index for time-series analysis. The dataset contained no missing values.

A 7-day moving average was used to smooth daily fluctuations and observe trends. Sales remained relatively stable throughout the year, showing no strong long-term drift.

Promotion Impact:
Average sales without promotion were approximately 200 units, while promotional days averaged around 248 units — a **23.9% increase**. This indicates promotions are the strongest sales driver.

Weekly Pattern:
Weekday sales outperformed weekends, with Wednesday as the best-performing day and Saturday as the weakest. This suggests an opportunity to strengthen weekend strategies.

Seasonality:
Monthly sales variation was minimal, indicating stable demand across the year.

3. Feature Engineering

To capture temporal patterns, several time-based features were engineered:

- **Lag Features:** lag_1, lag_7, lag_14 to capture sales momentum and weekly patterns.
- **Rolling Features:** 7-day and 30-day rolling averages to capture short- and long-term trends (shifted to prevent data leakage).
- **Calendar Features:** Month, DayOfMonth, and IsWeekend.
- **Interaction Feature:** Promo_Weekend to test promotional impact during weekends.

Rows with missing values from rolling calculations were removed before modeling.

4. Model Development

A **Gradient Boosting Regressor** was selected due to its strong performance on structured tabular data and its ability to capture non-linear relationships. Gradient Boosting builds decision trees sequentially, with each tree correcting errors from previous ones by minimizing residuals.

A chronological train-test split was used:

- First 306 days → Training set
- Last 30 days → Test set

Random splitting was avoided to preserve time order and prevent future data leakage.

The model was configured with 200 trees, maximum depth of 4, learning rate of 0.1, and subsampling to reduce overfitting. Feature importance analysis confirmed that **Promotion and DayOfWeek were the strongest predictors**, followed by lag and rolling features.

5. Model Evaluation & Forecasting

The model was evaluated on the final 30 days using:

- **MAE:** 12.95
- **RMSE:** 15.79
- **MAE as % of average sales:** 6.3%

An average prediction error of 6.3% indicates strong forecasting accuracy for retail sales.

To forecast January 1–7, 2025, a recursive multi-step approach was implemented. Each predicted value was appended to historical data so that lag and rolling features updated dynamically, simulating real-world deployment.

6. Business Insights

1. Promotions increase sales by nearly 24% and should be strategically optimized.
2. Weekend sales underperform, indicating opportunity for targeted campaigns.
3. Sales show strong short-term momentum, enabling reliable forecasting.
4. Demand is stable across months, supporting consistent inventory planning.
5. The model's low error rate makes it suitable for operational decision-making.

Conclusion

This project demonstrates an end-to-end time-series forecasting pipeline, including analysis, feature engineering, ensemble modelling, evaluation, and business interpretation. The Gradient Boosting model successfully captured key sales drivers and delivered accurate short-term forecasts, providing actionable insights for revenue optimization and planning.